

# Speed and velocity

Student: ..... Class: .....

## Calculating average speed

1. The average speed ( $v$ ) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

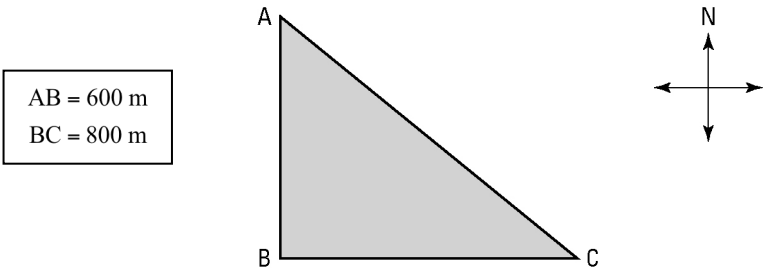
where  $d$  = distance travelled and  $t$  = time taken.

Complete the following table. Make sure you include the appropriate units for each quantity.

| Distance travelled | Time taken | Average speed |
|--------------------|------------|---------------|
| 100 m              | 20 s       |               |
| 3 km               |            | 60 km/h       |
|                    | 15 s       | 20 m/s        |
| 2 km               |            | 25 m/s        |

## Average speed and average velocity

2. Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right-angled triangle. A is north of B.



Both vehicles take 70 s for the trip.  
Perform the necessary calculations to complete the following table.

| Average speed of X | Average velocity of X | Average speed of Y | Average velocity of Y |
|--------------------|-----------------------|--------------------|-----------------------|
|                    |                       |                    |                       |